

Class 2

Geometric Features – GIS Representation Models

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A Day of Life with GIS

Activity	GIS Application
Alarm goes off	Electricity asset management
Shower	Hydraulic model predicts water usage
School transport	School bus route optimization
Lunch	Precision agriculture
Shopping	Target consumer marketing
Public participation	On-line collaboration
Train to Work	Real-Time Traffic Monitoring

GIS and Data Models

What is Geographic Information Systems?

What are Digital Spatial Datasets, Data models

The contents of the spatial database is a model of the earth. The essential function of spatial data we store and manipulate is to subdivide the earth's surface into entities or objects that can be characterized.

How do we describe geographical features /entities?

Spatial data

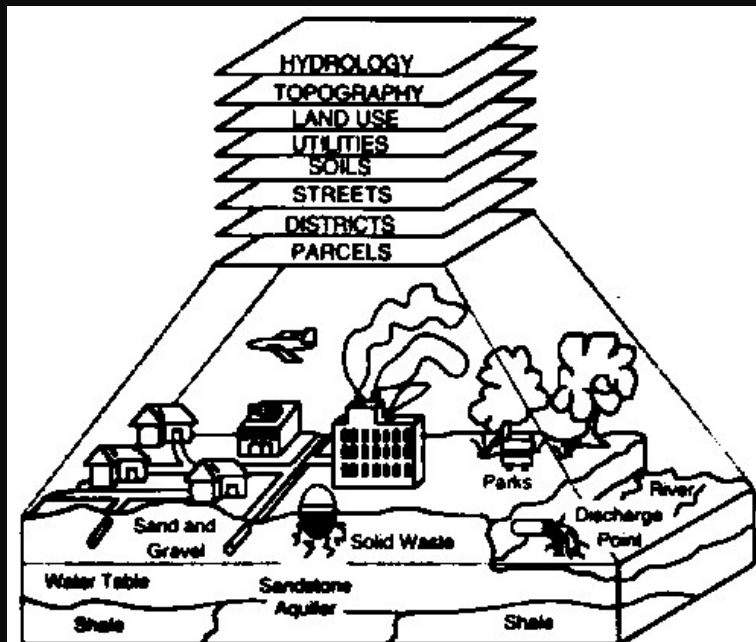
Non-spatial data (attribute)

Spatial Data Types

- continuous: elevation, rainfall, ocean salinity
- areas:
 - *unbounded*: land use, market areas, soils, rock type
 - *bounded*: city/county/state boundaries, ownership parcels, zoningnetworks: roads, transmission lines, streams
- points:
 - *fixed*: wells, street lamps,
 - *moving*: cars, fish, deer

How do we represent these digitally in a GIS?

Data layer or data plane based on similar characteristics
(e.g elevation, water lines, sewer lines)



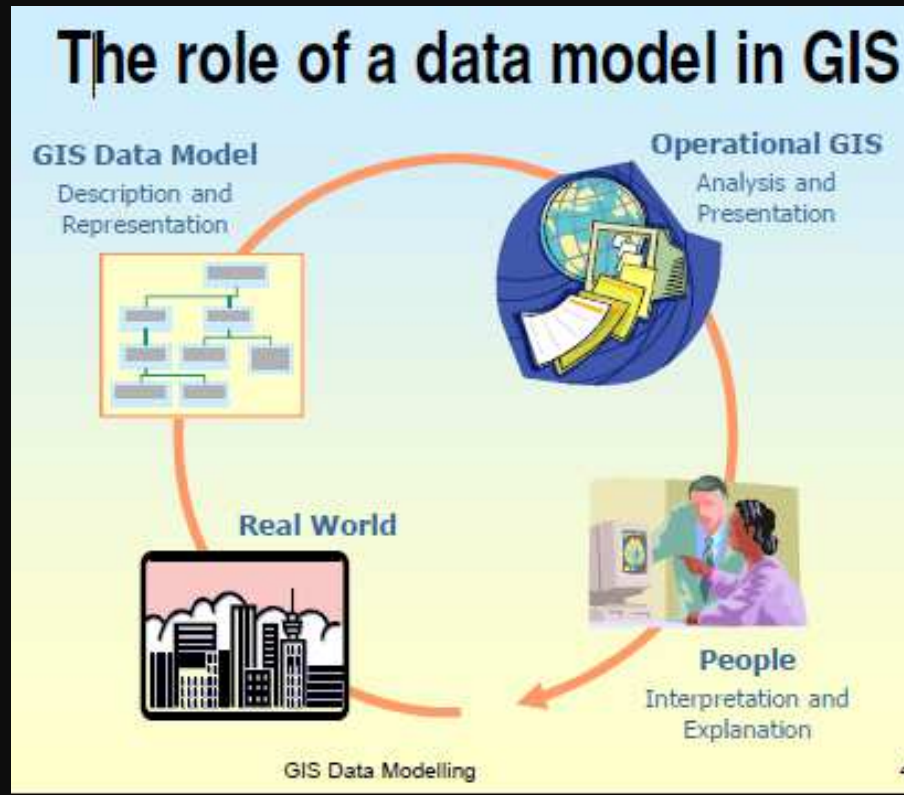
What is the Mode of representation?

Data models : vector (features) or
raster (surface) or TIN (Surface)
formats

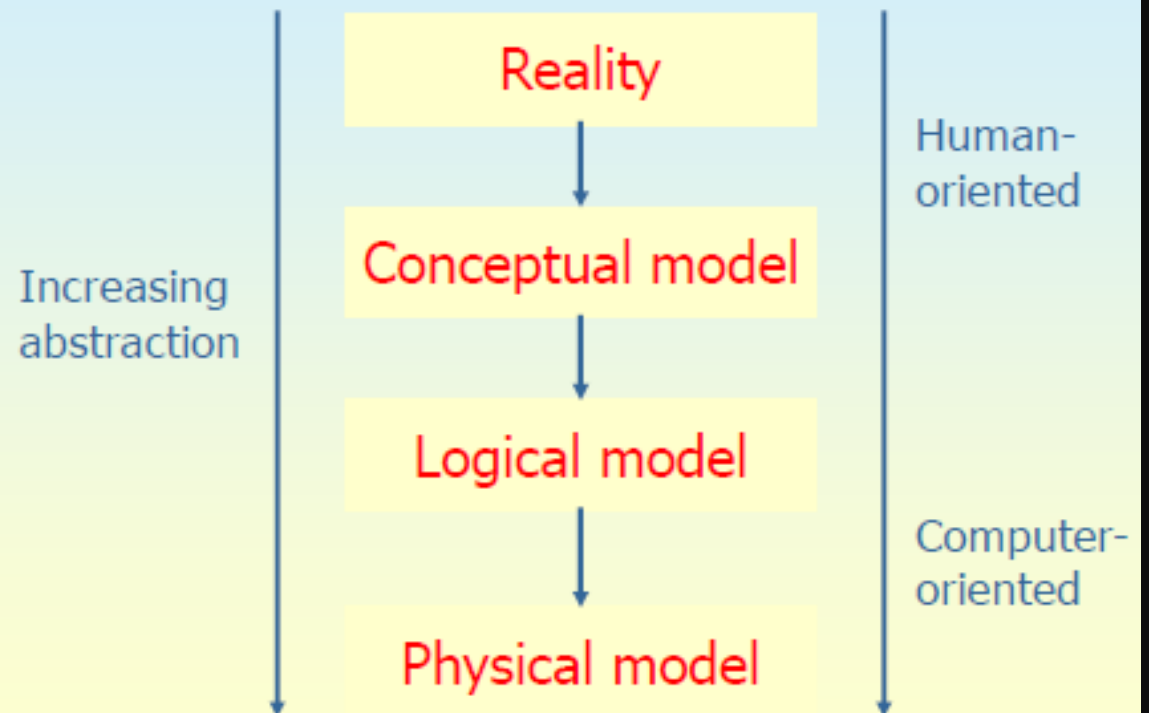
Choice of particular spatial data model
is important – why ?

What are the characteristics of an Ideal GIS Data Structure
Design & Purposes?

The GIS Data Model



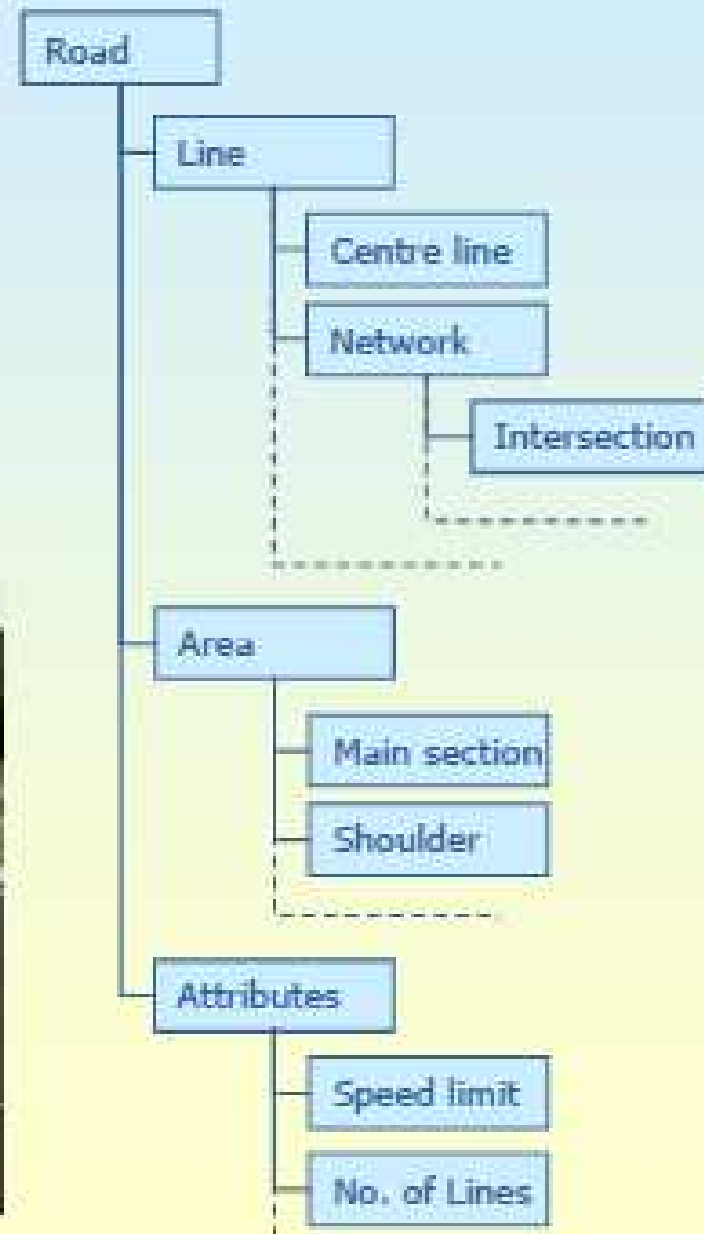
Levels of abstraction



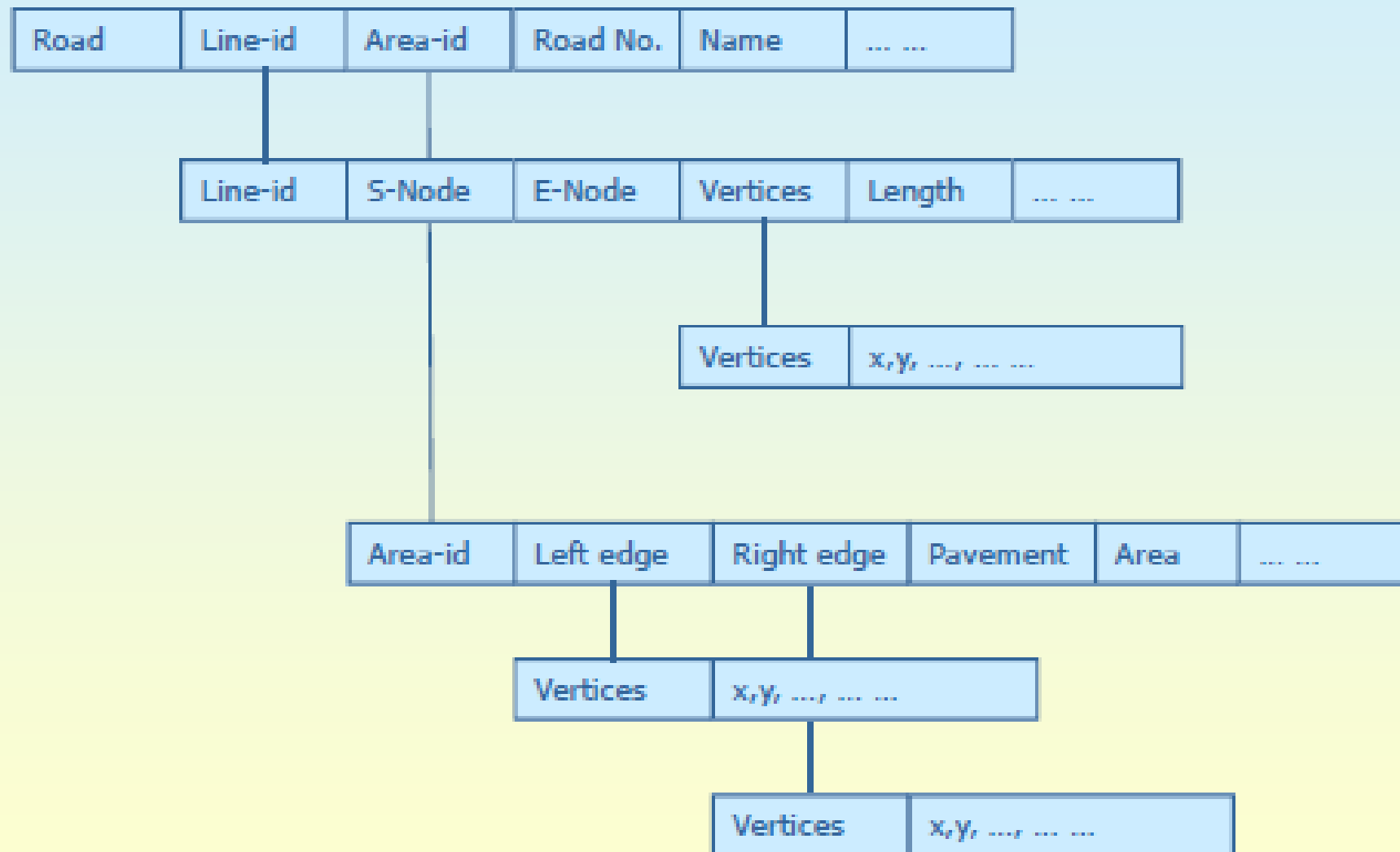
Conceptual model

A road:

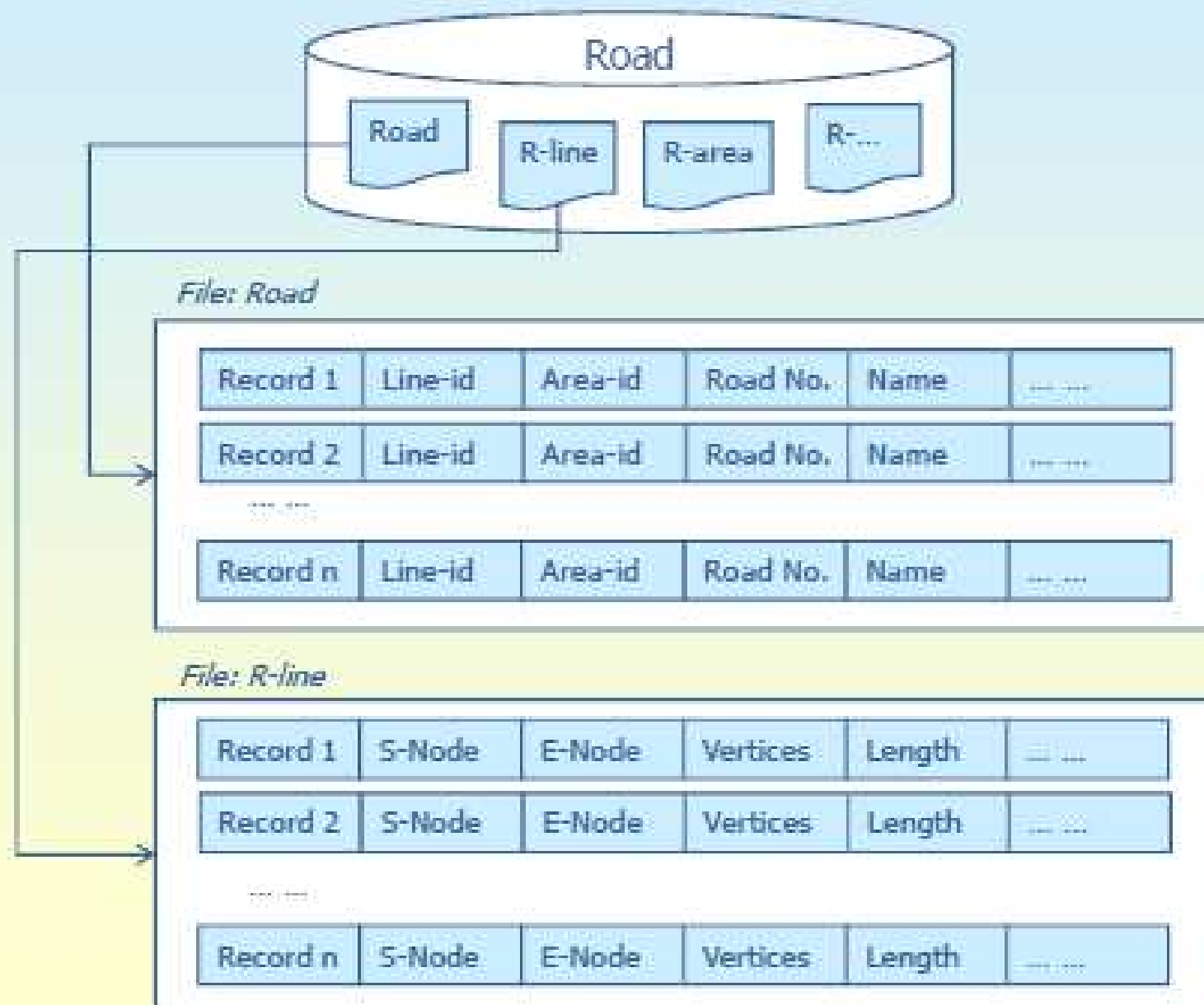
- Centre line (position)
- Edge line (width)
- Shoulder
- Number of lanes
- One way/two way
- Speed limit
- Traffic conditions
- Pavement
- Underground structure
- Intersections
- Date of completion
- Maintenance date
- ...



Logical model



Physical model



Types of Models

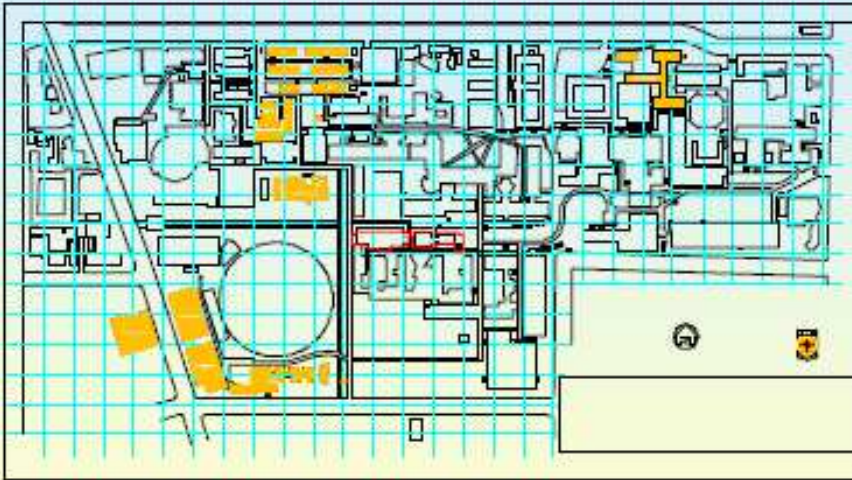
- Computer-Aided Design (CAD) – Engineering drawings
- Graphical (non-topological) - Simple maps without relationships.
- Image ; Raster/Grid - Grid of pixels (satellite images)
- Vector/Georelational Topological - Points, lines, polygons with relationships
- Network - Connected systems (roads, rivers).
- Triangulated Irregular Network (TIN) - 3D terrain using triangles.

Geographical data models used in GIS

Data model	Application
Computer-aided design	Simple mapping
Image	Image processing and simple grid analysis
Raster/grid	Spatial analysis and modelling
Vector/geo-relational topological	Operations on vector geometric features
Network	Network analysis
Triangulated irregular network (TIN)	Terrain analysis
Object	Operations on all types entities

Object - Smart geographic objects with properties and behavior.

The CAD data model



A CAD model focuses on feature drawing only so that it does not represent any kind of relationships between objects.

Simple graphical data model



A simple graphical data model is adequate to make a cartographic representation of Hong Kong SAR.

Image data model



Image data: On the left is the false colour composite of Landsat ETM image using Band 4, 3 and 2. On the right is the classification of the image using 6 bands of the Landsat ETM images (excluding Band 6).

Vector Data Models/Structures

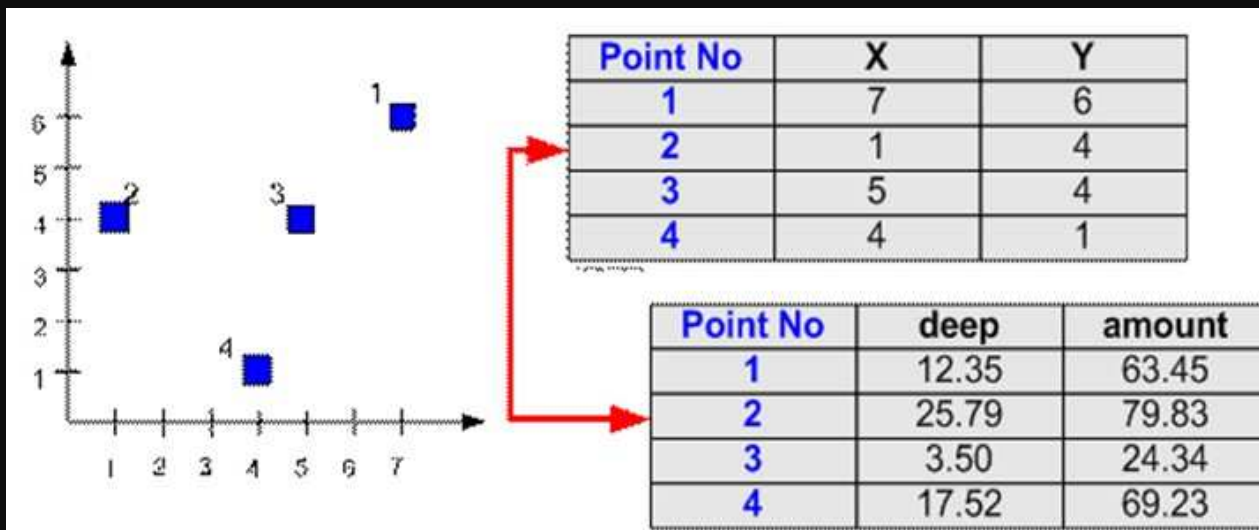
- One model for representing geographic space
- Spatial locations are explicit
- Relationships between entities/objects are implicit
- The vector data model provides for the precise positioning of features in space.
- Based on analytical geometry, a vector model builds a complex representation from primitive objects for the dimensions: points, lines and areas

In Summary- Vector described as a quantity with a starting coordinate and an associated displacement and direction.

Vector Data Models/Structures

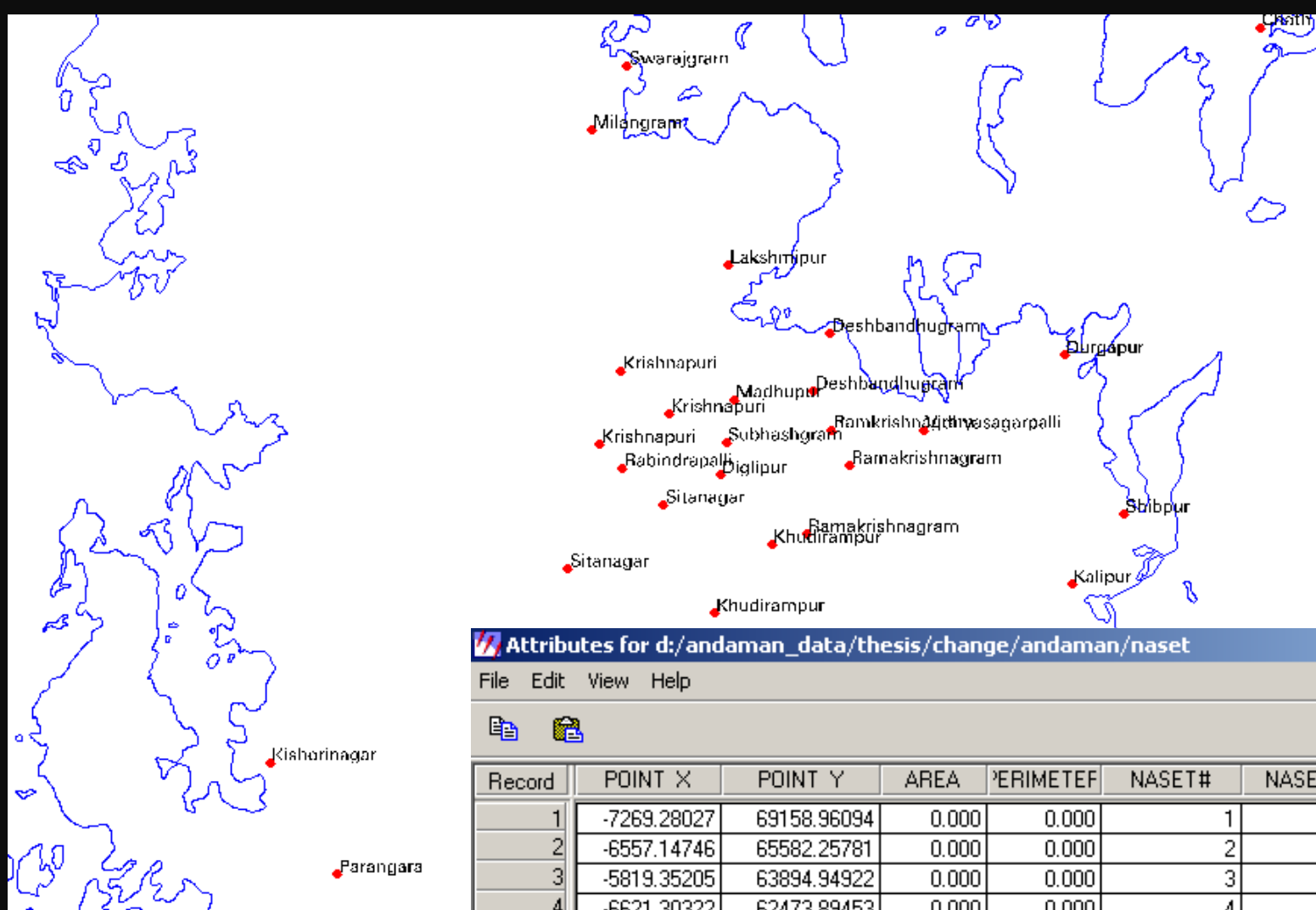
Points (0-dimension)

- Represent geographic features too small to be depicted as lines or areas (zero area)
- A point is recorded as a pair of x, y coordinates.
- **Node** is a topological point at which two or more arcs connect each other
- Each point is stored by its location (X, Y) together with the table attribute of this point.



Examples ?

Settlements



Attributes for d:/andaman_data/thesis/change/andaman/naset

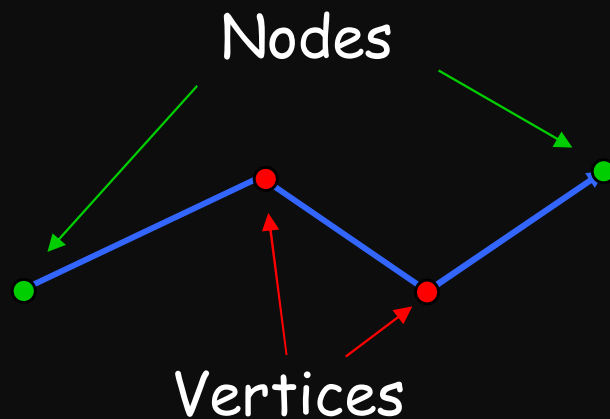
File Edit View Help

Record	POINT X	POINT Y	AREA	PERIMETER	NASET#	NASET-ID	NA SETNAM
1	-7269.28027	69158.96094	0.000	0.000	1	1	Radhanagar
2	-6557.14746	65582.25781	0.000	0.000	2	2	Swarajgram
3	-5819.35205	63894.94922	0.000	0.000	3	3	Swarajgram
4	-6621.30322	62473.89453	0.000	0.000	4	4	Milangram
5	-9280.57324	71837.47656	0.000	0.000	5	5	Shyamnagar
6	-7978.20508	70945.70313	0.000	0.000	6	6	Shyamnagar
7	-6643.75830	71657.83594	0.000	0.000	7	7	Shyamnagar
8	-8055.19238	70323.39063	0.000	0.000	8	8	Radhanagar
9	-9633.43164	72825.47656	0.000	0.000	9	9	Shyamnagar
10	-3579.31958	59497.56250	0.000	0.000	10	10	Lakshmipur
11	-1315.87598	57993.07813	0.000	0.000	11	11	Deshbandhugram
12	-6427.09326	55541.57422	0.000	0.000	12	12	Krishnapuri
13	-5956.94189	57173.67188	0.000	0.000	13	13	Krishnapuri
14	-4889.02686	56213.21875	0.000	0.000	14	14	Krishnapuri
15	-3444.99048	56535.60938	0.000	0.000	15	15	Madhupur

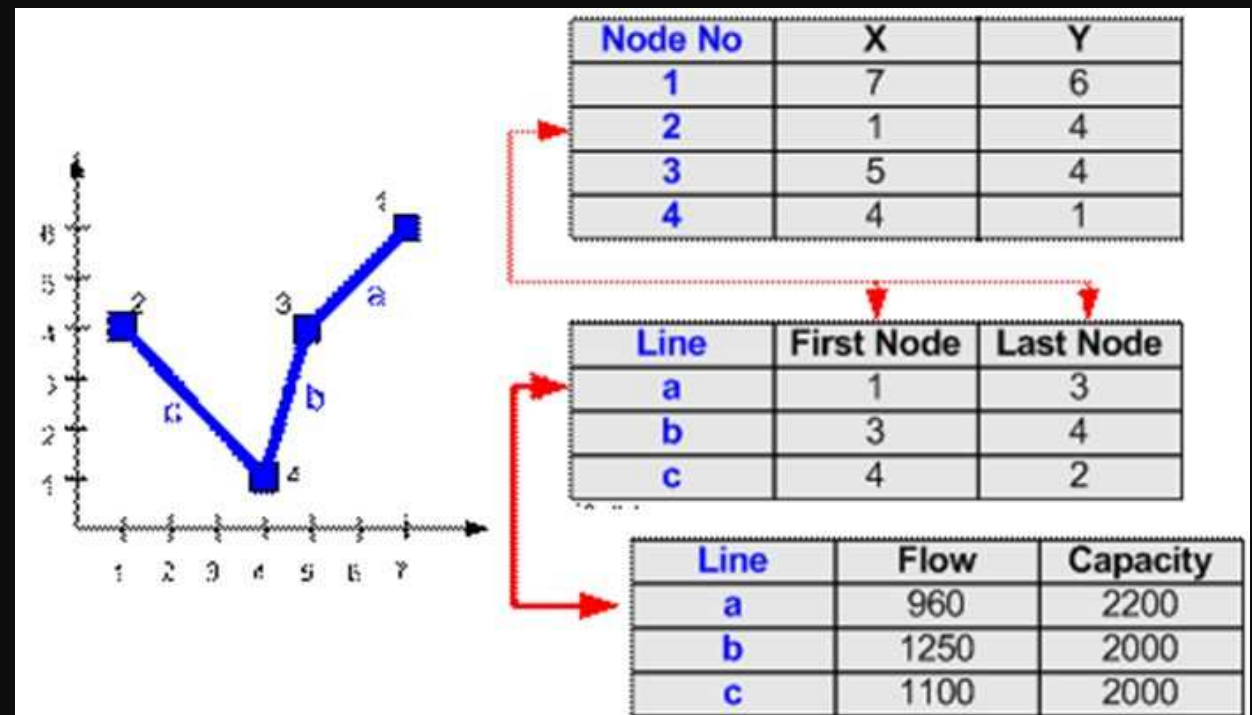
The Feature Attributes Are Displayed Here

Line/Arc (1-dimension)

- Represent geographic features too narrow to depict as areas;
- Lines are recorded as a series of ordered x,y coordinates.
- Each line is stored by the sequence of first and last point together with the associated table attribute of this line.



Examples ?



Drainage

Attributes for d:/andaman_data/thesis/topography/phytodiversity/drain

File Edit View Help

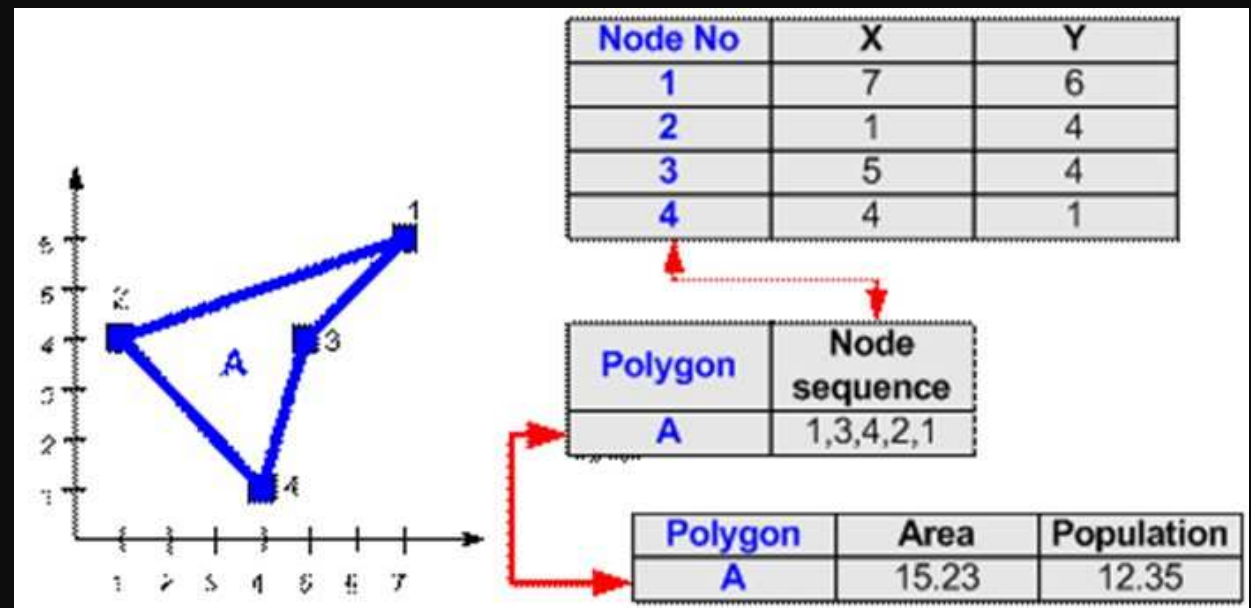
Record	FNODE#	TNODE#	LPOLY#	RPOLY#	LENGTH	DRAIN#	DRAIN-ID
1	7	12	1	1	112.722	1	57
2	14	9	1	1	111.616	2	129
3	17	12	1	1	392.089	3	129
4	24	16	1	1	356.342	4	57
5	22	24	1	1	112.022	5	57
6	18	23	1	1	455.246	6	82
7	28	14	1	1	1341.839	7	129
8	27	28	1	1	257.558	8	129
9	3	29	1	1	489.441	9	106
10	11	30	1	1	426.615	10	99
11	30	2	1	1	1104.619	11	149
12	31	30	1	1	79.008	12	99
13	32	22	1	1	394.235	13	58
14	36	20	1	1	349.188	14	104
15	38	34	1	1	346.194	15	58

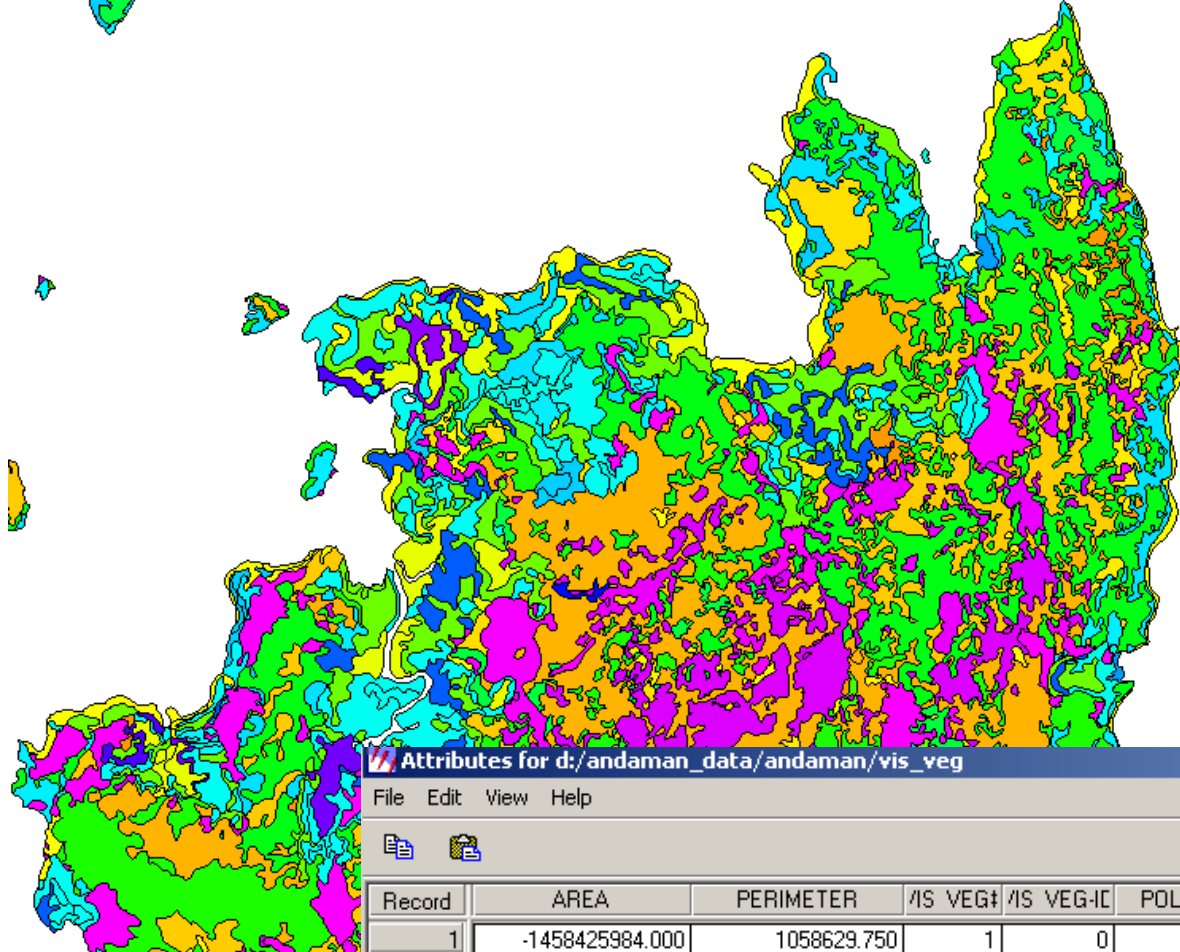
Polygon (2-dimensions)

- Represent homogeneous geographic features.
- Polygon is represented by a closed sequence of lines. Unlike line or poly-line (sequence of line), polygon always closed. That is, the first point is equal to the last point.

Examples ?

Using polygon, several geometric attributes such as area and perimeters can be derived easily





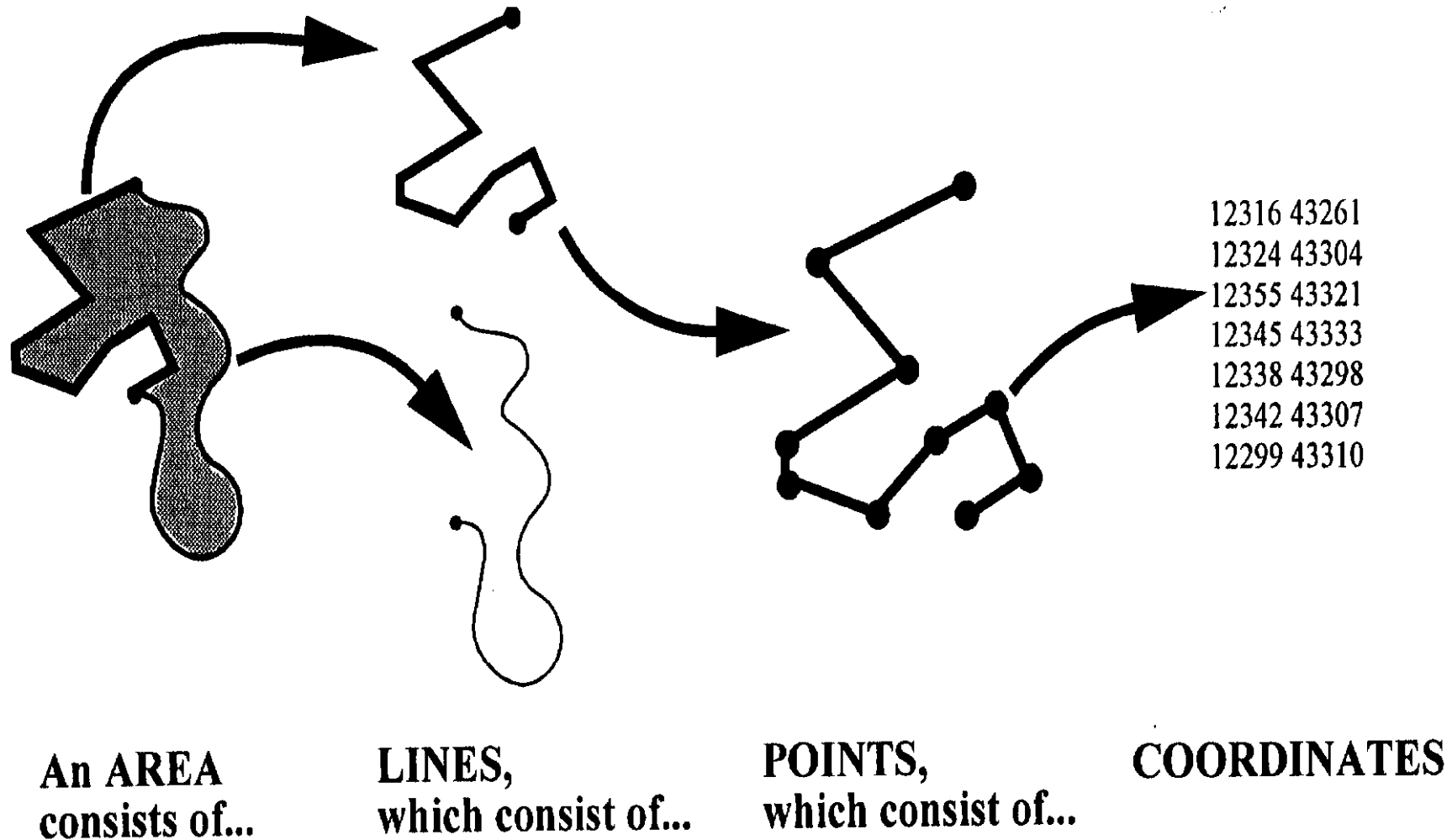
Forest Types

Attributes for d:/andaman_data/andaman/vis_veg

File Edit View Help

Record	AREA	PERIMETER	/IS VEG#	/IS VEG-IC	POLY#	BCLAS	LU CODE	CLASSNAME	TYPE	
1	-1458425984.000	1058629.750	1	0	0	0				
2	129172.328	2107.815	2	1	2	7213	06020300	MUD FLATS	MUD FLATS	MUD FLATS
3	2402721.250	15529.782	3	2	3	7212	03040602	MIXED MANGROVES	MIXED MANGROV	DENSE MANGROVE
4	71272.813	1211.564	4	3	4	7214	03040603	MIXED MANGROVES	MIXED MANGROV	DENSE MANGROVE
5	5256919.500	26790.762	5	4	5	7219	03041202	LITTORAL/BEACH FI	LITTORAL FORES	LITTORAL FOREST
6	119225.336	3733.029	6	5	6	7223	06020400	SAND	SAND	SAND
7	784066.313	5425.673	7	6	7	7211	06020300	MUD FLATS	MUD FLATS	MUD FLATS
8	40675.559	1426.787	8	7	8	1	03040600	MIXED MANGROVES	MIXED MANGROV	OPEN MANGROVE
9	30246.311	1211.823	9	8	9	2	06020400	SAND	SAND	SAND
10	107144.133	2955.620	10	9	10	3	03041202	LITTORAL/BEACH FI	LITTORAL FORES	LITTORAL FOREST
11	1234522.000	6959.299	11	10	11	7210	03040602	MIXED MANGROVES	MIXED MANGROV	DENSE MANGROVE
12	10732.756	496.730	12	12	13	5	03040600	MIXED MANGROVES	MIXED MANGROV	OPEN MANGROVE
13	13459.028	473.570	13	11	12	4	03041202	LITTORAL/BEACH FI	LITTORAL FORES	LITTORAL FOREST
14	50599.516	969.658	14	13	14	6	03041202	LITTORAL/BEACH FI	LITTORAL FORES	LITTORAL FOREST
15	35654.859	994.704	15	14	15	7	03040600	MIXED MANGROVES	MIXED MANGROV	OPEN MANGROVE
16	38900.160	1258.659	16	15	16	8	03040600	MIXED MANGROVES	MIXED MANGROV	OPEN MANGROVE
17	83852.461	1513.397	17	16	17	9	03041202	LITTORAL/BEACH FI	LITTORAL FORES	LITTORAL FOREST
18	723738.813	6014.719	18	17	18	10	03010902	SEMI-EVERGREEN E	SEMI-EVERGREE	SEMI-EVERGREEN
19	177905.750	2046.329	19	18	19	7217	06020300	MUD FLATS	MUD FLATS	MUD FLATS
20	98618.219	1799.120	20	19	20	7216	06020300	MUD FLATS	MUD FLATS	MUD FLATS

Feature Geometry



Geographic information has *dimension*. Areas are two-dimensional and consist of lines, which are one-dimensional and consist of points, which are zero-dimensional and consist of a coordinate pair.

Attribute data types

Within GIS the term attribute usually refers to records in a data table associated with individual features

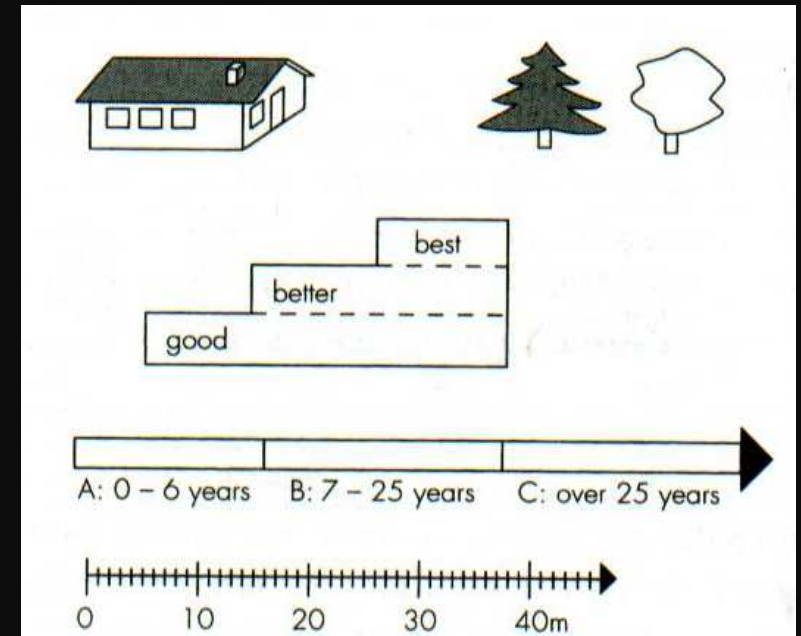
Two types –

Qualitative or nonnumeric
Quantitative or numeric

Qualitative further classified as –
Nominal and Ordinal

Quantitative as – Ratio and Interval

Cyclic – Wind direction, Slope, aspect

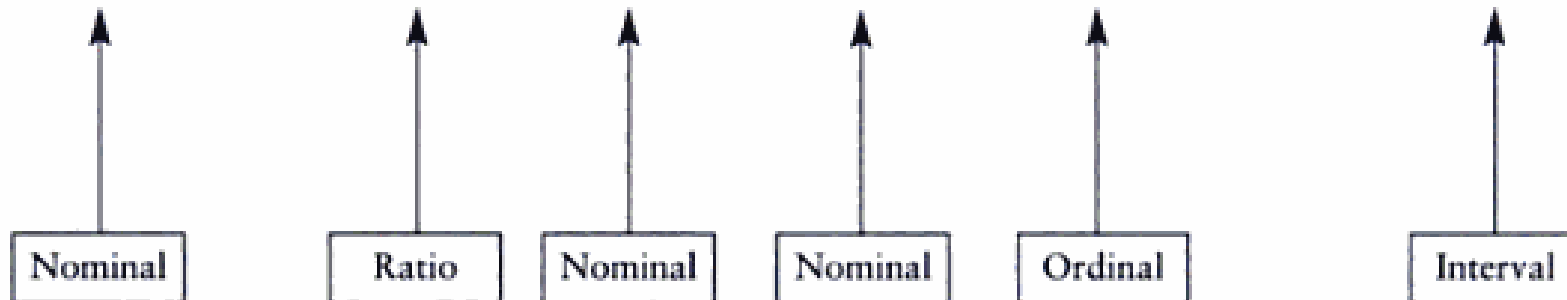


Source: Bernhardsen, Tor. (1999). 2nd Ed. *Geographic Information Systems: An Introduction*. pp 40.

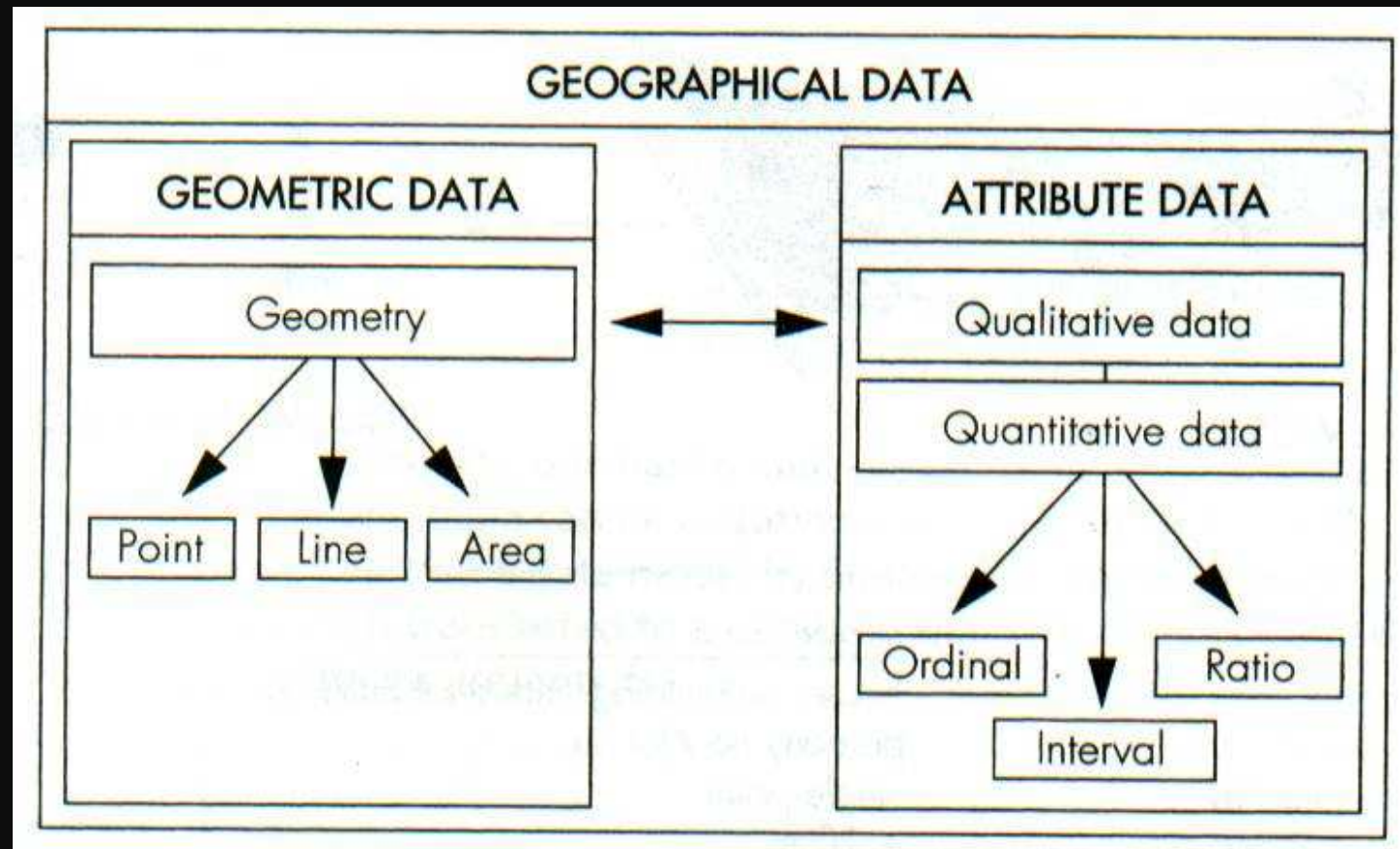
Example

Table 7.2 An example of a simple **data** table containing entity and **attribute data** collected about four individual study subjects. Columns are marked according to the level of measurement the **data** was collected at (**nominal**, **ordinal**, interval, or ratio)

<i>ID</i>	<i>Variable/Entity</i>	<i>Attribute_1</i>	<i>Attribute_2</i>	<i>Attribute_3</i>	<i>Attribute_4</i>	<i>...</i>	<i>Attribute_n</i>
1	Jane	53	Female	Hispanic	College	...	\$45,000
2	Chris	27	Male	Caucasian	High School	...	\$35,000
3	Bob	61	Male	Asian	College	...	\$55,000
4	Sue	36	Female	Caucasian	Graduate	...	\$65,000



Entity data and attribute data kept in separate files, perhaps a DBMS, which links them



Source: Bernhardsen, Tor. (1999). 2nd Ed. *Geographic Information Systems: An Introduction*. p. 40.

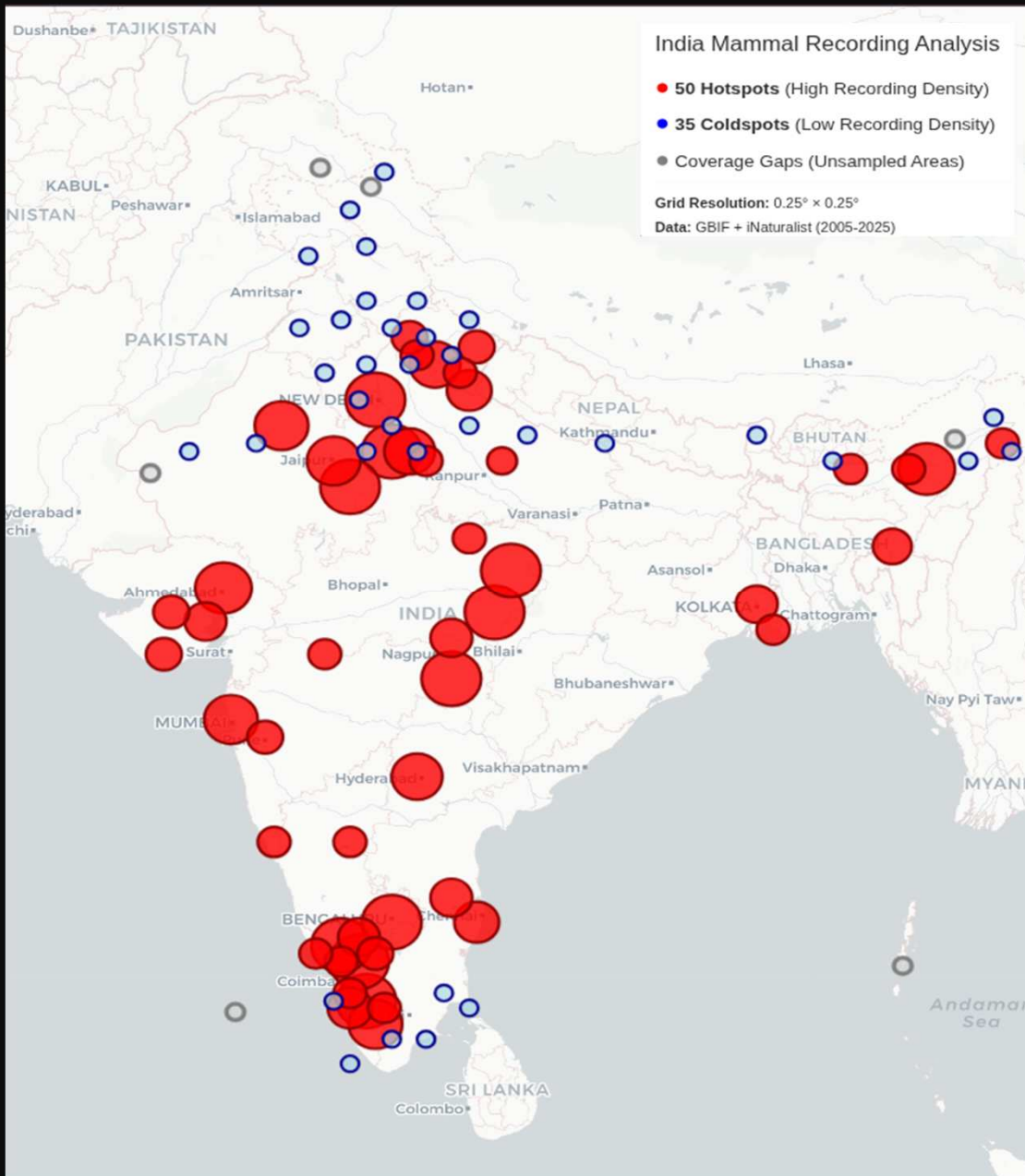
Data Symbolization

differences in:	symbols		
	point	line	area
size			
value			
grain			
colour			
orientation			
shape			

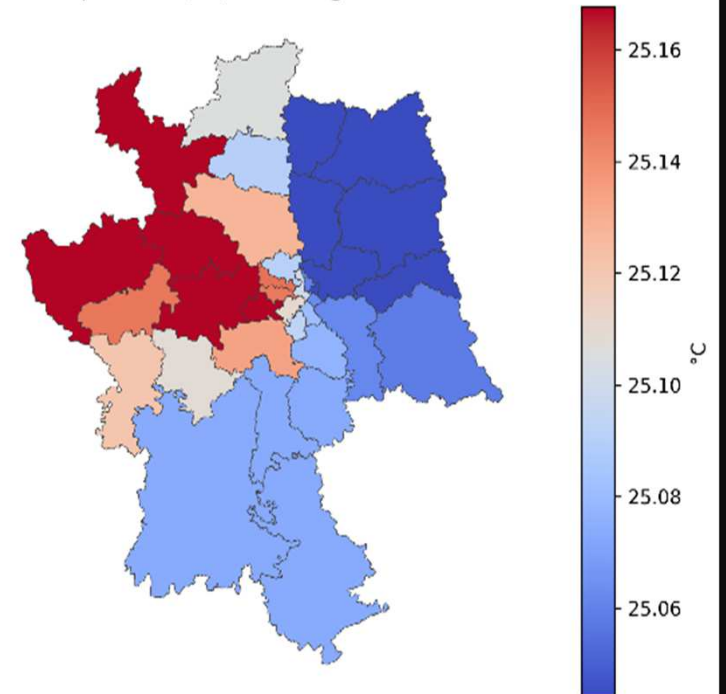
Plate 7

Representing spatial information: Bertin's (1993) graphical variables: size, value, texture (grain), orientation, and shape expressed in point, line, and area symbols.

(Source: Kraak and Ormeling 1996)

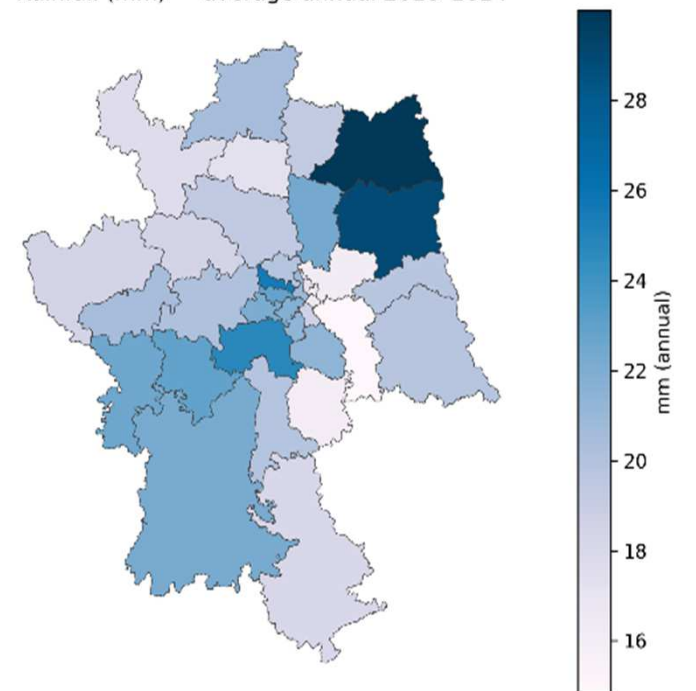


Temperature ($^\circ\text{C}$) — average 2018-2024



AQI — average 2018-2024

Rainfall (mm) — average annual 2018-2024



Exposure score — average 2018-2024